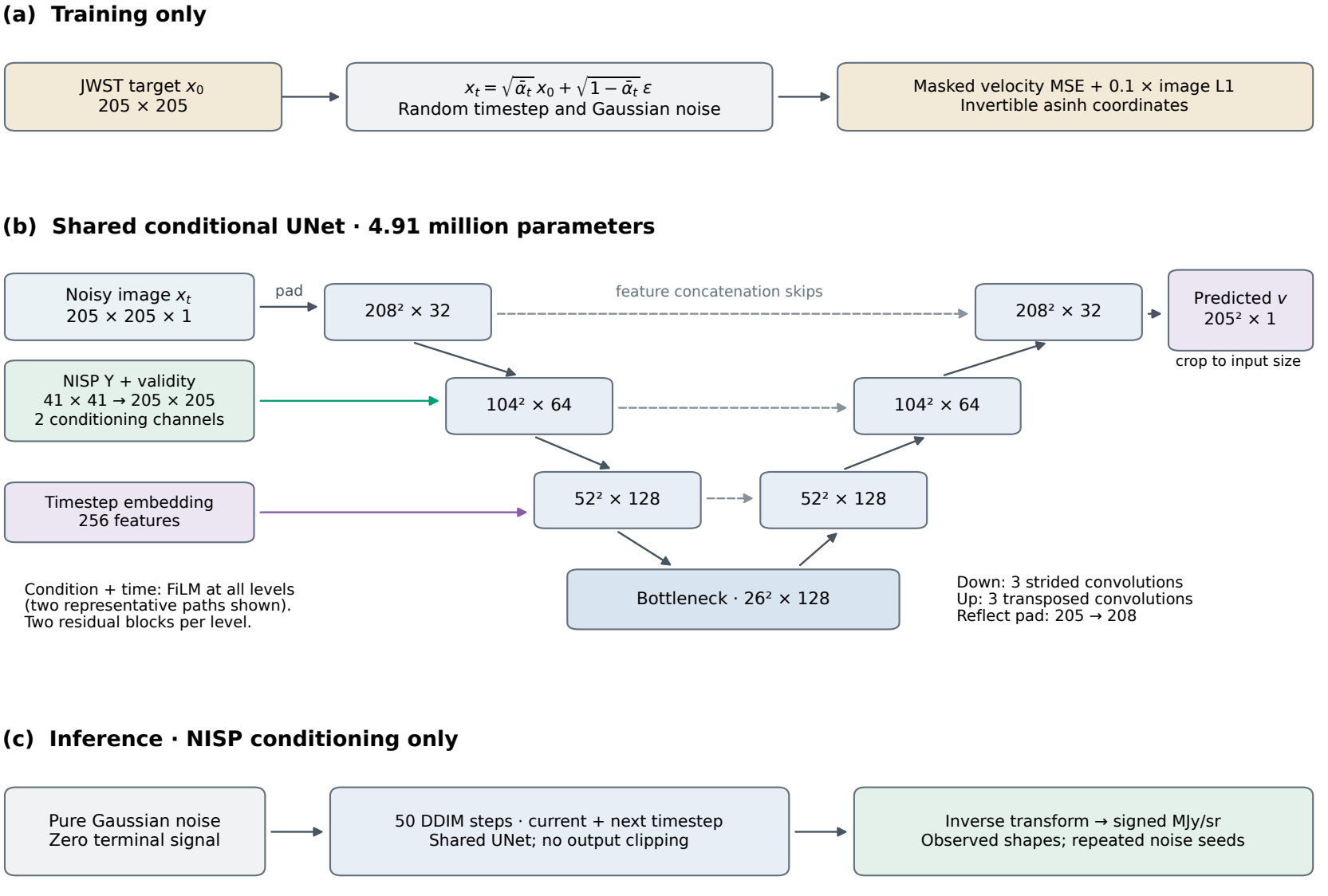
Conditional diffusion for NISP → F115W reconstruction
(a) Training only
JWST target x_0
205 × 205
$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$
Random timestep and Gaussian noise
Masked velocity MSE + 0.1 × image L1
Invertible asinh coordinates
(b) Shared conditional UNet · 4.91 million parameters
Noisy image x_t
205 × 205 × 1
pad
208² × 32
feature concatenation skips
208² × 32
Predicted v
205² × 1
crop to input size
NISP Y + validity
41 × 41 → 205 × 205
2 conditioning channels
104² × 64
104² × 64
Timestep embedding
256 features
52² × 128
52² × 128
Bottleneck · 26² × 128
Condition + time: FiLM at all levels
(two representative paths shown).
Two residual blocks per level.
Down: 3 strided convolutions
Up: 3 transposed convolutions
Reflect pad: 205 → 208
(c) Inference · NISP conditioning only
Pure Gaussian noise
Zero terminal signal
50 DDIM steps · current + next timestep
Shared UNet; no output clipping
Inverse transform → signed MJy/sr
Observed shapes; repeated noise seeds